



"FCL e-book": Table of Contents.

0. Introduction

1. Pre-requisites for Full Closed Loop

- 1.1 Well tuned hybrid closed loop
- 1.2 Fast insulin
- 1.3 Reliable insulin delivery from pump and cannula
- 1.4 Excellent CGM
- 1.5 Meal-related limitations?
- 1.6 Lifestyle-related limitations?
- 1.7 Time required for setting-up

Case study 1.1: Occlusion

Case study 1.2: Comparing insulins for FCL

Case study 1.3: Jumpy CGM

Case study 1.4: Lost pump connection

Case study 1.5: Permanent CGM values w/ 2x G6

Case study 1.6: When catastrophe strikes

Case study 1.7: MDI when the loop dies...

Case study 1.8: (placeholder for a Libre3 / 1.-minute case)...

...

2. General Settings for Full Closed Loop

- 2.1 SMB range extension
- 2.2 Max and min autoISF ratio
- 2.3 SMB delivery ratio
- 2.4 iobTH (iob_threshold_percent)
- 2.5 Eating Soon TT?
- 2.6 General settings in AAPS/Preferences

3. Description of autoISF / guidance by developers

- 3.1 Overview
- 3.2 ISF modulation flowcharts
- 3.3 Exercise mode and dynamic iobTH
- 3.4 Automation options with autoISF parameters
- 3.5 Activity monitor
- 3.6 Using one-minute CGM (Libre 3)
- 3.7 AutoISF parameters overview table
- 3.8 Emulator for logfile analysis and tuning
- 3.9 Links to related case studies/detailed doc.s

4. Meals: Setting ISF_weights in AAPS/Preferences

- 4.1 Getting started
- 4.2 bgAccel_ISF_weight
- 4.3 pp_ISF_weight
- 4.4 bgBrake_ISF_weight
- 4.5 dura_ISF and bg_ISF
- 4.6 Tuning your initial settings
- 4.7 Complex scenarios
- 4.8 Profile helper

Case study 4.1: Pizza

Case study 4.2: Low carb meals

Case study 4.3: Hands-off FCL around Christmas

5. Temp. modulation of autoISF aggressiveness

5.1 Automatic modulation of loop aggressiveness

- 5.1.1 autoISF off outside of meal windows

58	5.1.2 SMB off @ odd profile target
59	5.1.3 SMB off @ odd temp. target
60	5.1.4 Automatic diff. of FCL aggressiveness via Automations
61	5.1.5 Automatic diff. of FCL aggressiveness via Activity Monitor
62	5.1.6 Pro/con completely hands-off FCL
63	5.2 Manual modulation of FCL aggressiveness via DIY cockpit
64	5.2.1 Status recognition
65	5.2.2 Manual interventions from DIY cockpit
66	5.2.2.1 Temp. %profile or TT settings
67	5.2.2.2 Temp. settings in /preferences
68	5.2.2.3 Grey DIY cockpit buttons for pre-programmed FCL responses
69	5.2.2.4 Diff. of aggressiveness using "Automation State"
70	5.2.3 Temporary exit from FCL
71	5.3 Recognizing loop state from the AAPS home screen
72	5.3.1 Modulated loop aggressiveness (3 top buttons)
73	5.3.2 Color scheme of the top 3 buttons
74	5.3.3 Active Automations
75	5.3.4 FCL related indicator fields
76	5.3.5 Overall AAPS home screen
77	5.3.6 Extra graphs for autoISF
78	5.3.7 Info given every 5 minutes in the SMB tab
79	5.3.8 Info about last 15 autoISF decisions
80	5.3.9. SMB tab info when operating 1-minute Libre <i>(needs work)</i>
81	Summary: Your personal FCL cockpit
82	Case study 5.2: Sweet snacks / Glühwein w/ DIY cockpit
83	Case study 5.3: Compression low
84	
85	6. Temp. modulation for exercise and light (In-)activity
86	6.1 Dynamic iobTH and sensitivity ratio
87	6.1.1 Manual (direct) iobTH modulation
88	6.1.2 Automations for iobTH modulation
89	6.1.3 Dynamic iobTH
90	6.2 Temp. % profile switch
91	6.3 DIY cockpit based on User action Automations
92	6.4 Mastering the exercise after meal challenge
93	6.4.1 Manual mode
94	6.4.2 DIY cockpit button for User action Automation
95	6.5 Activity monitor based on step-counter
96	Case study 6.2 Biking day with hi carb lunch; DIY cockpit
97	
98	7. Advanced HCL (meal announcement via pre-bolus)
99	7.1 Hurdles for FCL
100	7.2 Getting ready to advance from HCL
101	7.3 Reduced pre-bolus
102	7.4 Tuning autoISF in HCL
103	7.5 Dealing with disturbances/ins. sens/resistance
104	7.6 Exercise management
105	7.7 Remote control (small children) <i>(fragment, to be completed)</i>
106	7.8 Other methods w/ meal announcement (MA)
107	Case study 7.1: Meal Announcement (5y old)
108	
109	8. Performance monitoring and tuning
110	Case study 8.2: Futility of tuning based on 1 extreme meal
111	
112	9. Trouble shooting
113	Case study 9.1: Un-intended loss of loop aggressiveness

114	
115	10. Emulator on PC to determine settings
116	10.1 Installing the Emulator on your PC
117	10.2 Analyzing loop decisions in logfiles
118	10.3 What-if analysis
119	
120	11. Emulator on the smartphone
121	11.1 Installing the emulator on your smartphone
122	11.2 Checking loop decisions on the smartphone
123	11.3 Options available on i-Phone (for Trio or iAPS, <i>placeholder</i>)
124	11.4 Real-time checking a „what-if“ question using speech synthesis
125	
126	12. Remarks for users of previous autoISF version
127	
128	13. Other avenues to Full Closed Loop
129	13.1 FCL using AAPS Master and Automations
130	Case study 13.1: Comparison: 1 month FCL, using Automation vs autoISF
131	13.2 dynamicISF used for Full Closed Loop
132	Case study 13.2: Using dynISF for FCL (<i>sorry, nobody provided interpretable data yet</i>)
133	13.3 Methods involving simple meal announcement that might be stretched into a FCL
134	13.3.1 Boost
135	Case study 13.3: Boost-based FCL for a child
136	13.3.2 AIMI,
137	13.3.3 EatingNow
138	13.3.4 Tsunami
139	13.4 No-bolus looping with precise carb inputs
140	13.5 Machine Learning (AI)
141	13.6 Dual hormone systems
142	13.7 Co-Medications that might facilitate FCL
143	Final Remark